

4.4 System timer, SysTick

When enabled, the system timer counts down from the reload value to zero, reloads (wraps to) the value in the SYST_RVR on the next clock cycle, then decrements on subsequent clock cycles. Writing a value of zero to the SYST_RVR disables the counter on the next wrap. When the counter transitions to zero, the COUNTFLAG status bit is set to 1. Reading SYST_CSR clears the COUNTFLAG bit to 0. Writing to the SYST_CVR clears the register and the COUNTFLAG status bit to 0. The write does not trigger the SysTick exception logic. Reading the register returns its value at the time it is accessed.

Note

When the processor is halted for debugging the counter does not decrement.

The system timer registers are:

Table 4-19 System timer registers summary

Address	Name	Type	Required privilege	Reset value	Description
0xE000E010	SYST_CSR	RW	Privileged	0x00000000	<i>SysTick Control and Status Register.</i>
0xE000E014	SYST_RVR	RW	Privileged	Unknown	<i>SysTick Reload Value Register on page 4-17.</i>
0xE000E018	SYST_CVR	RW	Privileged	Unknown	<i>SysTick Current Value Register on page 4-18.</i>
0xE000E01C	SYST_CALIB	RO	Privileged	IMPLEMENTATION DEFINED ^a	<i>SysTick Calibration Value Register on page 4-18.</i>

a. SysTick calibration value.

4.4.1 SysTick Control and Status Register

The SYST_CSR enables the SysTick features. See the register summary in [Table 4-19](#) for its attributes. The bit assignments are:

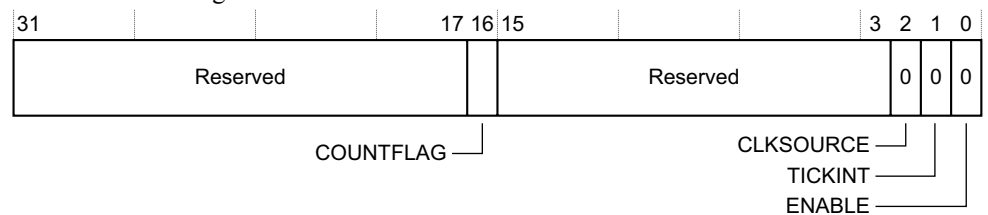


Table 4-20 SYST_CSR bit assignments

Bits	Name	Function
[31:17]	-	Reserved.
[16]	COUNTFLAG	Returns 1 if timer counted to 0 since the last read of this register.
[15:3]	-	Reserved.

Table 4-20 SYST CSR bit assignments (continued)

Bits	Name	Function
[2]	CLKSOURCE	Selects the SysTick timer clock source: 0 = external reference clock. 1 = processor clock.
[1]	TICKINT	Enables SysTick exception request: 0 = counting down to zero does not assert the SysTick exception request. 1 = counting down to zero asserts the SysTick exception request.
[0]	ENABLE	Enables the counter: 0 = counter disabled. 1 = counter enabled.

4.4.2 SysTick Reload Value Register

The SYST_RVR specifies the start value to load into the SYST_CVR. See the register summary in [Table 4-19 on page 4-16](#) for its attributes. The bit assignments are:

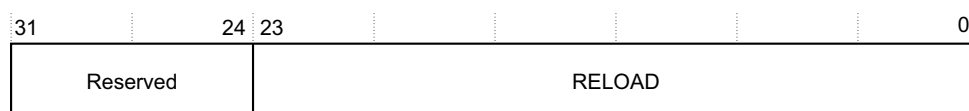


Table 4-21 SYST_RVR bit assignments

Bits	Name	Function
[31:24]	-	Reserved
[23:0]	RELOAD	Value to load into the SYST_CVR when the counter is enabled and when it reaches 0, see <i>Calculating the RELOAD value</i> .

Calculating the RELOAD value

The RELOAD value can be any value in the range 0x00000001-0x00FFFFFF. You can program a value of 0, but this has no effect because the SysTick exception request and COUNTFLAG are activated when counting from 1 to 0.

To generate a multi-shot timer with a period of N processor clock cycles, use a RELOAD value of N-1. For example, if the SysTick interrupt is required every 100 clock pulses, set RELOAD to 99.

